Exercise J-M

1. The number of real solutions of the equation $3\left(x^2+\frac{1}{x^2}\right)-2\left(x+\frac{1}{x}\right)+5=0$, is :

[JEE(Main)Jan.2023]

समीकरण $3\left(x^2+\frac{1}{x^2}\right)-2\left(x+\frac{1}{x}\right)+5=0$, के वास्तविक हलों की संख्या है : **[JEE(Main)Jan.2023]**

(A) 4 (B) 0 (C) 2 (D) 3

Ans. B

Sol. $3\left(x^2+\frac{1}{x^2}\right)-2\left(x+\frac{1}{x}\right)+5=0$

$3\left[\left(x+\frac{1}{x}\right)^2-2\right]-2\left(x+\frac{1}{x}\right)+5=0$

Let $x+\frac{1}{x}=t; t\in(-\infty-2]\cup[2,\infty)$

$3t^2-6-2t+5=0$

$3t^2-2t-1=0$

$t=1, t=\frac{-1}{3}$ But $t\in(-\infty,-2]\cup[2,\infty)$

$\Rightarrow$ No. real solution

2. Let $\lambda\neq0$ be a real number. Let α,β be the roots of the equation $14x^2-31x+3\lambda=0$ and α,γ be the roots of the equation $35x^2-53x+4\lambda=0$. Then $\frac{3\alpha}{\beta}$ and $\frac{4\alpha}{\gamma}$ are the roots of the equation : **[JEE(Main)Jan.2023]**

माना $\lambda\neq0$ एक वास्तविक संख्या है। माना α,β समीकरण $14x^2-31x+3\lambda=0$ के मूल हैं एवं α,γ समीकरण $35x^2-53x+4\lambda=0$ के मूल है। तब $\frac{3\alpha}{\beta}$ व $\frac{4\alpha}{\gamma}$ किस समीकरण के मूल है ? **[JEE(Main)Jan.2023]**

(A) $49x^2-245x+250=0$ (B) $7x^2+245x-250=0$

(C) $49x^2+245x+250=0$ (D) $7x^2-245x+250=0$

Ans. A

Sol. $14\alpha^2-31\alpha+3\lambda=0$ ...(i)

$35\alpha^2-53\alpha+4\lambda=0$ ...(ii)

$\because$ α will satisfy both give equations

$\therefore\ 14\alpha^2-31\alpha+3\lambda=0\Rightarrow\alpha^2-\frac{31\alpha}{14}+\frac{3\lambda}{14}=0$ .... (iii)

$35\alpha^2 - 53\alpha + 4\lambda = 0 \Rightarrow \alpha^2 - \frac{53\alpha}{35} + \frac{4\lambda}{35} = 0$ ....(iv)

from (iii) – (iv) we get

$\lambda = 7\alpha$ put in ... (iii)

$$\alpha^2 - \frac{31\alpha}{14} + \frac{3}{14}(7\alpha) = 0$$

$$\alpha^2 - \frac{31\alpha}{14} + \frac{3\alpha}{2} = 0$$

$$\Rightarrow \alpha = 0 \text{ or } \alpha = \frac{5}{7}$$

($\alpha = 0$ not acceptable as $\lambda \neq 0$)

$$\therefore \alpha = \frac{5}{7}$$

$$\because \alpha + \beta = \frac{31}{14} \Rightarrow \beta = \frac{3}{2}$$

$$\because \alpha + \gamma = \frac{53}{35} \Rightarrow \gamma = \frac{4}{5}$$

$$\therefore \frac{3\alpha}{\beta} = \frac{10}{7} \text{ and } \frac{4\alpha}{\gamma} = \frac{25}{7}$$

$$\therefore \text{ Reqd. Q.E. is } \left(x - \frac{10}{7}\right)\left(x - \frac{25}{7}\right) = 0$$

$(7x - 10)(7x - 25) = 0$

$49x^2 - 245x + 250 = 0$

3. The number of real roots of the equation $\sqrt{x^2 - 4x + 3} + \sqrt{x^2 - 9} = \sqrt{4x^2 - 14x + 6}$, is :

[JEE(Main)Jan.2023]

समीकरण $\sqrt{x^2 - 4x + 3} + \sqrt{x^2 - 9} = \sqrt{4x^2 - 14x + 6}$ के वास्तविक मूलों की संख्या है :

[JEE(Main)Jan.2023]

(A) 3 (B) 0 (C) 1 (D) 2

Ans. C

Sol. $\sqrt{(x-1)(x-3)} + \sqrt{(x-3)(x+3)} = \sqrt{2(2x-1)(x-3)}$

$x - 3 = 0$ and $\sqrt{x-1} + \sqrt{x+3} = \sqrt{2(2x-1)}$ .....(1)

Squaring both side; $x - 1 + x + 3 + 2\sqrt{(x-1)(x+3)} = 4x - 2$

$2\sqrt{(x-1)(x+3)} = 2x - 4$

$\sqrt{x^2 + 3x - x - 3} = x - 2$ ..... (2)

Squaring both side; $x^2 + 2x - 3 = x^2 - 4x + 4$

$6x = 7 \Rightarrow x = \frac{7}{6}$

At $x = \frac{7}{6}$ equation (2) positive = negative

Rejected

$\therefore$ real root $x = 3$

4. The equation $e^{4x} + 8e^{3x} + 13e^{2x} - 8e^{x} + 1 = 0, x \in R$ has : **[JEE(Main)Jan.2023]**

(A) four solutions two of which are negative

(B) two solutions and only one of them is negative

(C) two solutions and both are negative

(D) no solution

समीकरण $e^{4x} + 8e^{3x} + 13e^{2x} - 8e^{x} + 1 = 0, x \in R$: **[JEE(Main)Jan.2023]**

(A) के चार हल हैं जिनमें से दो ऋणात्मक है

(B) के दो हल हैं तथा उनमें से केवल एक ऋणात्मक है

(C) के दो हल हैं जिनमें से दो ऋणात्मक है

(D) का कोई हल नहीं है

Ans. C

Sol. $e^{4x} + 8e^{3x} + 13e^{2x} - 8e^{x} + 1 = 0$

Let $e^{x} = t$

Now $t^4 + 8t^3 + 13t^2 - 8t + 1 = 0$

Dividing equation by t^2,

$$t^2 + 8t + 13 - \frac{8}{t} + \frac{1}{t^2} = 0$$

$$t^2 + \frac{1}{t^2} + 8\left(t - \frac{1}{t}\right) + 13 = 0$$

$$\left(t - \frac{1}{t}\right)^2 + 2 + 8\left(t - \frac{1}{t}\right) + 13 = 0$$

Let $t - \frac{1}{t} = z$

$z^2 + 8z + 15 = 0$

$(z+3)(z+5) = 0$

$z = -3$ ro $z = -5$

So, $t - \frac{1}{t} = -3$ or $t - \frac{1}{t} = -5$

$t^2 + 3t - 1 = 0$ or $t^2 + 5t - 1 = 0$

$t = \frac{-3 \pm \sqrt{13}}{2}$ or $t = \frac{-5 \pm \sqrt{29}}{2}$

as $t = e^x$ so t must be positive,

$t = \frac{\sqrt{13}-3}{2}$ or $\frac{\sqrt{29}-5}{2}$

So, $x = \ln\left(\frac{\sqrt{13}-3}{2}\right)$ or $x = \ln\left(\frac{\sqrt{29}-5}{2}\right)$

Hence two solution and both are negative.

5. Let $f:R-\{2,6\} \to R$ be real valued function defined as $f(x) = \frac{x^2+2x+1}{x^2-8x+12}$. Then range of f is

[JEE(Main)Jan.2023]

माना $f:R-\{2,6\} \to R$ एक वास्तविक मान फलन है, जो $f(x) = \frac{x^2+2x+1}{x^2-8x+12}$ द्वारा परिभाषित है। तो f का परिसर है।

[JEE(Main)Jan.2023]

(A) $\left(-\infty,-\frac{21}{4}\right)\cup\left[\frac{21}{4},\infty\right)$ (B) $\left(-\infty,-\frac{21}{4}\right)\cup[1,\infty)$

(C) $\left(-\infty,-\frac{21}{4}\right]\cup[0,\infty)$ (D) $\left(-\infty,-\frac{21}{4}\right)\cup(0,\infty)$

Ans. C

Sol. Let $y = \frac{x^2+2x+1}{x^2-8x+12}$

By cross multiplying

$yx^2 - 8xy + 12y - x^2 - 2x - 1 = 0$

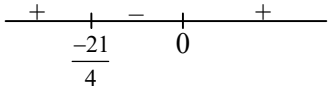
$x^2(y-1) - x(8y+2) + (12y-1) = 0$

Case 1, $y \neq 1$

$D \geq 0$

$\Rightarrow (8y+2)^2 - 4(y-1)(12y-1) \geq 0$

$\Rightarrow y(4y+21) \geq 0$



$y \in \left(-\infty,\frac{-21}{4}\right]\cup[0,\infty)-\{1\}$

Case 2, $y = 1$

$x^2 + 2x + 1 = x^2 - 8x + 12$

$10x = 11$

$x = \frac{11}{10}$ So, y can be 1

Hence $y \in \left(-\infty, \frac{-21}{4}\right] \cup [0, \infty)$

6. The number of integral values of k, for which one root of the equation $2x^2 - 8x + k = 0$ lies in the interval (1,2) and its other roots lies in the interval (2,3), is : **[JEE(Main)Jan.2023]**

k के पूर्णांक मानों, जिनके लिए समीकरण $2x^2 - 8x + k = 0$ का एक मूल अंतराल (1,2) में है, तथा दूसरा मूल अंतराल (2,3) में, की संख्या है : **[JEE(Main)Jan.2023]**

(A) 0 (B) 3 (C) 1 (D) 2

Ans. C

Sol. Let $f(x) = 2x^2 - 8x + k = 0$

$f(1) > 0$

$f(2) < 0$

$f(3) > 0$

$f(1) > 0 \Rightarrow 2 - 8 + k > 0 \Rightarrow k > 6$ (A)

$f(2) < 0 \Rightarrow 8 - 16 + k < 0 \Rightarrow k < 8$ (B)

$f(3) > 0 \Rightarrow 18 - 24 + k > 0 \Rightarrow k > 6$ (C)

$k \in (6, 8) \therefore$ integral $k = 7$

7. Let $\alpha_1, \alpha_2, ..., \alpha_7$ be the roots of the equation $x^7 + 3x^5 - 13x^3 - 15x = 0$ and $|\alpha_1| \geq |\alpha_2| \geq ... \geq |\alpha_7|$. Then $\alpha_1\alpha_2 - \alpha_3\alpha_4 + \alpha_5\alpha_6$ is equal to _____ . **[JEE(Main)Jan.2023]**

माना समीकरण $x^7 + 3x^5 - 13x^3 - 15x = 0$ के मूल $\alpha_1, \alpha_2, ..., \alpha_7$ हैं तथा $|\alpha_1| \geq |\alpha_2| \geq ... \geq |\alpha_7|$. है। तो $\alpha_1\alpha_2 - \alpha_3\alpha_4 + \alpha_5\alpha_6$ बराबर है। **[JEE(Main)Jan.2023]**

Ans. 9

Sol. $x^7 + 3x^5 - 13x^3 - 15x = 0$

$\Rightarrow$ $x = 0, x^6 + 3x^4 - 13x^2 - 15 = 0$

Let $x^2 = t$

Equation $t^3 + 3t^2 - 13t - 15 = 0$

$\Rightarrow (t - 3)(t^2 + 6t + 5) = 0$

$\Rightarrow t = 3, t^2 + 6t + 5 = 0$

$\Rightarrow t = -1, -5$

$x^2 = 3 \Rightarrow x = \sqrt{3}, -\sqrt{3}$

$x^2 = -1 \Rightarrow x = i, -i$

$x^2 = -5 \Rightarrow x = -\sqrt{5}i, \sqrt{5}i$

$a_1a_2 - a_3a_4 + a_5a_6$

$=\left(\sqrt{5}i\right)\left(-\sqrt{5}i\right)-\left(\sqrt{3}\right)\left(-\sqrt{3}\right)+(i)(-i)$

$=5+3+1=9$

8. Let $a \in R$ and let α,β be the roots of the equation $x^2+60^{\frac{1}{4}}x+a=0$. If $\alpha^4+\beta^4=-30$, then the product of all possile values of a is _____. **[JEE(Main)Jan.2023]**

माना $a \in R$ तथा समीकरण $x^2+60^{\frac{1}{4}}x+a=0$ के मूल α,β है। यदि $\alpha^4+\beta^4=-30$ है, तो a के सभी संभव मानों का गुणनफल है_____. **[JEE(Main)Jan.2023]**

Ans. 45

Sol. Since, $\alpha^4+\beta^4=-30$

$\Rightarrow\left[(\alpha+\beta)^2-2\alpha\beta\right]^2-2\alpha^2\beta^2=-30$ $\Rightarrow\left[\left(-60^{1/4}\right)^2-2a\right]^2-2a^2=-30$

$\Rightarrow(\sqrt{60}-2a)^2=2a^2-30$ $\Rightarrow 60+4a^2-4a\sqrt{60}=2a^2-30$

$\Rightarrow 2a^2-4a(\sqrt{60})+90=0$ $\Rightarrow a^2-2a(\sqrt{60})+45=0$

So, product of value of a = 45

Product = 45

9. If the value of real number $a>0$ for which $x^2 - 5ax + 1 = 0$ and $x^2 - ax - 5 = 0$ have a common real root is $\frac{3}{\sqrt{2\beta}}$ then β is equal to _____. **[JEE(Main)Jan.2023]**

यदि वास्तविक संख्या $a>0$ जिसके लिए $x^2 - 5ax + 1 = 0$ तथा $x^2 - ax - 5 = 0$ का उभयनिष्ठ वास्तविक मूल है, का मान $\frac{3}{\sqrt{2\beta}}$ है, तब β बराबर है। **[JEE(Main)Jan.2023]**

Ans. 13

Sol. Let α be the common root

so $\alpha^2 - 5a\alpha + 1 = 0$

$\alpha^2 - a\alpha - 5 = 0$

$\Rightarrow\frac{\alpha^2}{26a}=\frac{\alpha}{6}=\frac{1}{4a}$

$\Rightarrow\alpha^2=\frac{13}{2}\Rightarrow\alpha=\sqrt{\frac{13}{2}}\Rightarrow a=\frac{3}{2\alpha}=\frac{3}{\sqrt{2\beta}}$

$\Rightarrow\beta=13$

10. The sum of all the roots of the equation $\left|x^2-8x+15\right|-2x+7=0$ is : **[JEE(Main)April,2023]**

समीकरण $\left|x^2-8x+15\right|-2x+7=0$ के सभी मूलों का योग है : **[JEE(Main)April,2023]**

(A) $11+\sqrt{3}$ (B) $11-\sqrt{3}$ (C) $9+\sqrt{3}$ (D) $9-\sqrt{3}$

Ans. C

Sol. $\left|x^2-8x+15\right|-2x+7=0$

$|(x-3)(x-5)|-2x+7=0$

Case – I : If $x \in (-\infty,3]\cup[5,\infty)$

$x^2-8x+15-2x+7=0$

$x^2-10x+22=0$

$x=\frac{10\pm\sqrt{12}}{2}=5\pm\sqrt{3}$

$\therefore x=5+\sqrt{3}$ is accepted

Case – II : If $x \in (3,5)$

$-x^2+8x-15-2x+7=0$

$x^2-6x+8=0$

$x=2,4 \Rightarrow x=4$ is only accepted

$\therefore$ Sum of roots $5+\sqrt{3}+4=9+\sqrt{3}$

11. Let α, β, γ be the three roots of the equation $x^3+bx+c=0$. if $\beta\gamma=1=-\alpha$, then $b^3+2c^3-3\alpha^3-6\beta^3-8\gamma^3$ is equal to: **[JEE(Main)April,2023]**

माना α, β, γ समीकरण $x^3+bx+c=0$ के तीन मूल है। यदि $\beta\gamma=1=-\alpha$, तो $b^3+2c^3-3\alpha^3-6\beta^3-8\gamma^3$ बराबर है। **[JEE(Main)April,2023]**

(A) 21 (B) 19 (C) $\frac{169}{8}$ (D) $\frac{155}{8}$

Ans. B

Sol. Roots of $x^3+bx+c=0$ are α,β,γ

$\because \beta\gamma=1=-\alpha$

$\therefore \alpha=-1$ ....(i)

and $\alpha+\beta+\gamma=0$ ....(ii)

$\alpha\beta\gamma=-c$ ....(iii)

$\therefore c=1$ ....(iv)

$\beta+\gamma=1$ ....(v)

$\because \alpha\beta+\beta\gamma+\gamma\alpha=b$

$\Rightarrow \alpha(\beta+\gamma)+\beta\gamma=b$

$\therefore b=0$ ....(vi)

$\therefore \beta=-\omega,\gamma=-\omega^2$ ....(vii)

$b^3+2c^3-3\alpha^3-6\beta^3-8\gamma^3$

$=0+2+3+6+8$

$=19$

12. Let α, β be the roots of the quadratic equation $x^2+\sqrt{6}x+3=0$. Then $\frac{\alpha^{23}+\beta^{23}+\alpha^{14}+\beta^{14}}{\alpha^{15}+\beta^{15}+\alpha^{10}+\beta^{10}}$ is equal to : **[JEE(Main)April,2023]**

माना द्विघातीय समीकरण $x^2+\sqrt{6}x+3=0$ के मूल α, β हैं। तो $\frac{\alpha^{23}+\beta^{23}+\alpha^{14}+\beta^{14}}{\alpha^{15}+\beta^{15}+\alpha^{10}+\beta^{10}}$ बराबर है : **[JEE(Main)April,2023]**

(A) 72 (B) 81 (C) 729 (D) 9

Ans. D

Sol. $\frac{\alpha^{23}+\beta^{23}+\alpha^{14}+\beta^{14}}{\alpha^{15}+\beta^{15}+\alpha^{10}+\beta^{10}}$

Let $a_n=\alpha^n+\beta^n$

$=\frac{a_{23}+a_{14}}{a_{15}+a_{10}}$

$x^2+\sqrt{6}x+3=0 \begin{cases} \alpha \\ \beta \end{cases}$

$x^2+\sqrt{6}x+3=0 \sum_{\beta}^{\alpha}$

$x=\frac{-\sqrt{6}\pm\sqrt{-6}}{2}$

$=\sqrt{6}\left(\frac{-1\pm i}{2}\right)$

$=\sqrt{3}\left(\frac{-1\pm i}{\sqrt{2}}\right)$

$\alpha=\sqrt{3}e^{\frac{i3\pi}{4}}$

13. The number of elements in the set $\left\{n\in\mathbb{Z}:\left|n^2-10n+19\right|<6\right\}$ is _____. **[JEE(Main)April,2023]**

समुच्चय $\left\{n\in\mathbb{Z}:\left|n^2-10n+19\right|<6\right\}$ में अवयवों की संख्या है _____ **[JEE(Main)April,2023]**

Ans. 6

Sol. $-6<n^2-10n+19<6$

$\Rightarrow n^2-10n+25>0$ and $n^2-10n+13<0$

$(n-5)^2 > 0$ $\quad 5-3\sqrt{2} < n < 5+3\sqrt{2}$

$N \in Z - \{5\}$ $\quad n = \{2,3,4,5,6,7,8\}$

.....(i) $\quad$ ....(ii)

From (i) $\cap$ (ii)

N = {2, 3, 4, 5, 6, 8,}

Number of values of n = 6

14. If a and b are the roots of the equation $x^2 - 7x - 1 = 0$, then the value of $\frac{a^{21} + b^{21} + a^{17} + b^{17}}{a^{19} + b^{19}}$ is equal to _____.

[JEE(Main)April,2023]

यदि समीकरण $x^2 - 7x - 1 = 0$ के मूल a तथा b हैं, तो $\frac{a^{21} + b^{21} + a^{17} + b^{17}}{a^{19} + b^{19}}$ का मान बराबर है।

[JEE(Main)April,2023]

Ans. 51

Sol. $x^2 - 7x - 1 = 0 <_b^a$

By newton's theorem

$S_{n+2} - 7S_{n+1} - S_n = 0$

$S_{21} - 7S_{20} - S_{19} = 0$

$S_{20} - 7S_{19} - S_{18} = 0$

$S_{19} - 7S_{18} - S_{17} = 0$

$$\frac{S_{21} + S_{17}}{S_{19}} = \frac{S_{21} + (S_{19} - 7S_{18})}{S_{19}}$$

$$= \frac{S_{21} + S_{19} - 7(S_{20} - 7S_{19})}{S_{19}}$$

$$= \frac{50S_{19} + (S_{21} - 7S_{20})}{S_{19}}$$

$$= 51 \cdot \frac{S_{19}}{S_{19}} = 51$$

15. If α, β are the roots of the equation $x^2 - x - 1 = 0$ and $S_n = 2023\alpha^n + 2024\beta^n$, then :

[JEE(Main)Jan.2024]

यदि समीकरण $x^2 - x - 1 = 0$ के मूल α, β हैं तथा $S_n = 2023\alpha^n + 2024\beta^n$, तो : **[JEE(Main)Jan.2024]**

(A) $S_{11} = S_{10} + S_{12}$ (B) $2S_{12} = S_{11} + S_{10}$ (C) $S_{12} = S_{11} + S_{10}$ (D) $2S_{11} = S_{12} + S_{10}$

Ans. C

Sol. We have, $x^2 - x - 1 = 0$

By Newton's method

$S_n - S_{n-1} - S_{n-2} = 0$

$\therefore S_n = S_{n-1} + S_{n-2}$

$\therefore S_{12} = S_{11} + S_{10}$

16. Let S be the set of positive integral values of a for which $\frac{ax^2 + 2(a+1)x + 9a + 4}{x^2 - 8x + 32} < 0, \forall x \in R$. Then, the number of elements in S is : **[JEE(Main)Jan.2024]**

माना a के धन पूर्णांक मानों, जिन के लिए $\frac{ax^2 + 2(a+1)x + 9a + 4}{x^2 - 8x + 32} < 0, \forall x \in R$ है, का समुच्चय S है। तो S में अवयवों की संख्या है : **[JEE(Main)Jan.2024]**

(A) 3 (B) 0 (C) ∞ (D) 1

Ans. B

Sol. $ax^2 + 2(a + 1)x + 9a + 4 < 0 \ \forall \ x \in R$

$\therefore a < 0$

17. Let $S = \left\{x \in R : \left(\sqrt{3} + \sqrt{2}\right)^x + \left(\sqrt{3} - \sqrt{2}\right)^x = 10\right\}$. Then the number of elements in S is : **[JEE(Main)Jan.2024]**

माना $S = \left\{x \in R : \left(\sqrt{3} + \sqrt{2}\right)^x + \left(\sqrt{3} - \sqrt{2}\right)^x = 10\right\}$ है। तो S में अवयवों की संख्या है : **[JEE(Main)Jan.2024]**

(A) 4 (B) 0 (C) 1 (D) 2

Ans. D

Sol. $\left(\sqrt{3} + \sqrt{2}\right)^x + \left(\sqrt{3} - \sqrt{2}\right)^x = 10$

Let $\left(\sqrt{3} + \sqrt{2}\right)^x = t$

$t + \frac{1}{t} = 10$

$t^2 - 10t + 1 = 0$

$t = \frac{10 \pm \sqrt{100 - 4}}{2} = 5 \pm 2\sqrt{6}$

$\left(\sqrt{3} + \sqrt{2}\right)^x = \left(\sqrt{3} \pm \sqrt{2}\right)^2$

$x = 2$ or $x = -2$

Number of solutions = 2

18. Let $\alpha, \beta \in N$ be roots of the equation $x^2 - 70x + \lambda = 0$, where $\frac{\lambda}{2}, \frac{\lambda}{3} \notin N$. If λ assumes the minimum possible value, then $\frac{\left(\sqrt{\alpha-1}+\sqrt{\beta-1}\right)(\lambda+35)}{|\alpha-\beta|}$ is equal to _____. **[JEE(Main)Jan.2024]**

माना समीकरण $x^2 - 70x + \lambda = 0$, जहाँ $\frac{\lambda}{2}, \frac{\lambda}{3} \notin N$ हैं, के मूल $\alpha, \beta \in N$ हैं। λ के न्यूनतम संभव मानों के लिए, $\frac{\left(\sqrt{\alpha-1}+\sqrt{\beta-1}\right)(\lambda+35)}{|\alpha-\beta|}$ बराबर है _____। **[JEE(Main)Jan.2024]**

Ans. 60

Sol. $x^2 - 7x + \lambda = 0$

$\alpha + \beta = 70$

$\alpha\beta = \lambda$

$\therefore \alpha(70 - \alpha) = \lambda$

Since, 2 and 3 does not divide λ

$\therefore \alpha = 5, \beta = 65, \lambda = 325$

By putting value of α, β, λ we get the required value of 60.

19. Let the sum of the maximum and the minimum values of the function $f(x) = \frac{2x^2 - 3x + 8}{2x^2 + 3x + 8}$ be $\frac{m}{n}$, where gcd(m, n) = 1. Then m + n is equal to : **[JEE(Main)April,2024]**

माना फलन $f(x) = \frac{2x^2 - 3x + 8}{2x^2 + 3x + 8}$ के उच्चतम तथा न्यूनतम मानों का योग $\frac{m}{n}$, जहाँ gcd(m, n) = 1, है। तो m + n बराबर है : **[JEE(Main)April,2024]**

(A) 182 (B) 217 (C) 195 (D) 201

Ans. D

Sol. $f(x) = \frac{2x^2 - 3x + 8}{2x^2 + 3x + 8} = y, 2x^2 + 3x + 8 > 0 \quad \forall x \in R$

$\Rightarrow x^2(2y - 2) + x(3y + 3) + 8y - 8 = 0$

Since $x \in R$, the equation has real roots

$\Rightarrow D \geq 0$

$\Rightarrow (3y + 3)^2 - 4(2y - 2)(8y - 8) \geq 0$

$\Rightarrow 9(y + 1)^2 - 64y(y - 1)^2 > 0$

$\Rightarrow (3y+3)^2-(8y-8)^2 \geq 0$

$\Rightarrow (11y-5)(-5y+11) \geq 0$

$\Rightarrow \left(y-\frac{5}{11}\right)\left(y-\frac{11}{5}\right) \leq 0$

$\Rightarrow y \in \left[\frac{5}{11}, \frac{11}{5}\right]$

Sum of maximum and minimum value

$y_{max}+y_{min}=\frac{5}{11}+\frac{11}{5}=\frac{25+121}{55}$

$=\frac{146}{55}=\frac{m}{n} \Rightarrow m+n=201$

20. If 2 and 6 are the roots of the equation $ax^2+bx+1=0$, then the quadratic equation, whose roots are $\frac{1}{2a+b}$ and $\frac{1}{6a+b}$, is : **[JEE(Main)April,2024]**

यदि समीकरण $ax^2+bx+1=0$ के मूल 2 तथा 6 हैं, तो वह द्विघात समीकरण, जिसके मूल $\frac{1}{2a+b}$ तथा $\frac{1}{6a+b}$ हैं, है : **[JEE(Main)April,2024]**

(A) $2x^2+11x+12=0$ (B) $x^2+8x+12=0$ (C) $4x^2+14x+12=0$ (D) $x^2+10x+16=0$

Ans. B

Sol. $\text{Sum}=8=-\frac{b}{a}$

$\text{Product}=12=\frac{1}{a} \qquad \Rightarrow a=\frac{1}{12}$

$b=-\frac{2}{3}$

$2a+b=\frac{2}{12}-\frac{2}{3}=-\frac{1}{2}$

$6a+b=\frac{6}{12}-\frac{2}{3}=-\frac{1}{6}$

sum = –8

$x^2+8x+12=0$

21. Let α, β be the distinct roots of the equation $x^2 - \left(t^2 - 5t + 6\right)x + 1 = 0, t \in \mathbb{R}$ and $a_n = \alpha^n + \beta^n$. Then the minimum value of $\frac{a_{2023} + a_{2025}}{a_{2024}}$ is : **[JEE(Main)April,2024]**

यदि α, β समीकरण $x^2 - \left(t^2 - 5t + 6\right)x + 1 = 0, t \in \mathbb{R}$ के असमान मूल हैं तथा $a_n = \alpha^n + \beta^n$ है। तो $\frac{a_{2023} + a_{2025}}{a_{2024}}$ का न्यूनतम मान है : **[JEE(Main)April,2024]**

(A) –1/2 (B) 1/2 (C) 1/4 (D) –1/4

Ans. D

Sol. From Newton's formula

$\alpha_{n+2} - (t^2 - 5t + 6)\,\alpha_{n+1} + \alpha_n = 0$

$n = 2023$

$\alpha_{2025} - (t^2 5t + 6)\,\alpha_{2024} + \alpha_{2023} = 0$

$$\frac{\alpha_{2025} + \alpha_{2023}}{\alpha_{2024}} = t^2 - 5t + 6$$

$$= \frac{-D}{4a} = -\frac{1}{4}$$

22. Let α, β be the roots of the equation $x^2 + 2\sqrt{2}x - 1 = 0$. Then quadratic equation, whose roots are $\alpha^4 + \beta^4$ and $\frac{1}{10}\left(\alpha^6 + \beta^6\right)$, is : **[JEE(Main)April,2024]**

माना समीकरण $x^2 + 2\sqrt{2}x - 1 = 0$ के मूल α, β है। द्विघात समीकरण, जिसके मूल $\alpha^4 + \beta^4$ तथा $\frac{1}{10}\left(\alpha^6 + \beta^6\right)$ हैं, है : **[JEE(Main)April,2024]**

(A) $x^2 - 180x + 9506 = 0$ (B) $x^2 - 195x + 9466 = 0$

(C) $x^2 - 195x + 9506 = 0$ (D) $x^2 - 190x + 9466 = 0$

Ans. C

Sol. $x^2 + 2\sqrt{2}x - 1 = 0$

$\alpha + \beta = -2\sqrt{2}$

$\alpha\beta = -1$

$\alpha^4 + \beta^4 = (\alpha^2 + \beta^2)^2 - 2\alpha^2\beta^2$

$= ((\alpha + \beta)^2 - 2\alpha\beta)^2 - (\alpha\beta)^2$

$= (8 + 2)^2 - 2\,(-1)^2$

$= 100 - 2 = 98$

$\alpha^6 + \beta^6 = ((\alpha + \beta)\,((\alpha + \beta)^2 - 3\,\alpha\beta))^2 - 2(\alpha\beta)^3$

$= \left(-2\sqrt{2}\,(8+3)\right)^2 + 2$

$= (8)\,(121) + 2 = 970$

$\frac{1}{10}\left(\alpha^6 + \beta^6\right) = 97$

$x^2 - (98 + 97)x + (98)\,(97) = 0$

$\Rightarrow x^2 - 195x + 95x + 9506 = 0$

23. Let α, β; $\alpha > \beta$, be the roots of the equation $x^2 - \sqrt{2}x - \sqrt{3} = 0$. Let $P_n = \alpha^n - \beta^n$, $n \in N$. Then $\left(11\sqrt{3} - 10\sqrt{2}\right)P_{10} + \left(11\sqrt{2} + 10\right)P_{11} - 11P_{12}$ is equal to : **[JEE(Main)April,2024]**

माना समीकरण $x^2 - \sqrt{2}x - \sqrt{3} = 0$ के मूल α, β; $\alpha > \beta$ हैं। माना $P_n = \alpha^n - \beta^n$, $n \in N$. है। तो $\left(11\sqrt{3} - 10\sqrt{2}\right)P_{10} + \left(11\sqrt{2} + 10\right)P_{11} - 11P_{12}$ बराबर है : **[JEE(Main)April,2024]**

(A) $11\sqrt{2}P_9$ (B) $11\sqrt{3}P_9$ (C) $10\sqrt{3}P_9$ (D) $10\sqrt{2}P_9$

Ans. C

Sol. $x^2 - \sqrt{2}x - \sqrt{3} = 0$

$P_n = \alpha^n - \beta^n$

α and β are the roots of the equation

Using Newton's theorem

$P_{n+2} - \sqrt{2}P_{n+1} - \sqrt{3}P_n = 0$

Put $n = 10$

$P_{12} - \sqrt{2}P_{11} - \sqrt{3}P_{10} = 0$

$P_{12} = \sqrt{2}P_{11} + \sqrt{3}P_{10}$

Put $n = 9$

$P_{11} - \sqrt{2}P_{10} - \sqrt{3}P_9 = 0$

$P_{11} = \sqrt{2}P_{10} + \sqrt{3}P_9$

$\left(11\sqrt{3} - 10\sqrt{2}\right)P_{10} + \left(11\sqrt{2} + 10\right)P_{11} - 11P_{12}$

Put the value of P_{12} and P_{12} in above equation.

$= \left(11\sqrt{3} - 10\sqrt{2}\right)P_{10} + \left(11\sqrt{2} + 10\right)\left(\sqrt{2}P_{10} + \sqrt{3}P_9\right) - 11\left(\sqrt{2}P_{11} + \sqrt{3}P_{10}\right)$

$$= 11\sqrt{3}P_{10} - 10\sqrt{2}P_{10} + 22P_{10} + 10\sqrt{2}P_{10} + 11\sqrt{6}P_9 + 10\sqrt{3}P_9 - 11\sqrt{2}P_{11} - 11\sqrt{3}P_{10}$$

$$= 22P_{10} + 11\sqrt{6}P_9 + 10\sqrt{3}P_9 - 11\sqrt{2}\left(\sqrt{2}P_{10} + \sqrt{3}P_9\right)$$

$$= 22P_{10} + 11\sqrt{6}P_9 + 10\sqrt{3}P_9 - 22P_{10} - 11\sqrt{6}P_9$$

$$= 10\sqrt{3}P_9$$

24. Let α, β be roots of $x^2 + \sqrt{2}x - 8 = 0$. If $U_n = \alpha^n + \beta^n$, then $\frac{U_{10} + \sqrt{2}U_9}{2U_8}$ is equal to_____.

[JEE(Main)April,2024]

माना $x^2 + \sqrt{2}x - 8 = 0$ के मूल α, β हैं। यदि $U_n = \alpha^n + \beta^n$, तो $\frac{U_{10} + \sqrt{2}U_9}{2U_8}$ बराबर है_____

[JEE(Main)April,2024]

Ans. 4

Sol. $x^2 + \sqrt{2} - 8 = 0 \overset{\alpha}{\underset{\beta}{\nearrow\searrow}} \Rightarrow \alpha^2 + \sqrt{2}\alpha = 8$

$$\alpha + \beta = \sqrt{2},\ \alpha\beta = -8 \quad \Rightarrow \alpha + \sqrt{2} = \frac{8}{\alpha}$$

$$\Rightarrow \beta + \sqrt{2} = \frac{8}{\beta}$$

$$\frac{U_{10} + \sqrt{2}U_9}{2U_8} = \frac{\alpha^{10} + \beta^{10} + \sqrt{2}\alpha^9 + \sqrt{2}\beta^9}{2\alpha^8 + 2\beta^8}$$

$$= \frac{\alpha^9\left(\alpha + \sqrt{2}\right) + \beta^9\left(\beta + \sqrt{2}\right)}{2\left(\alpha^8 + \beta^8\right)}$$

$$= \frac{\alpha^9.\left(\frac{8}{\alpha}\right) + \beta^9\left(\frac{8}{\beta}\right)}{2\left(\alpha^8 + \beta^8\right)} = \frac{8}{2} = 4$$